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Chapter 26: A Brief History of the Discrete Leviathan (Part I)—Collecting You (1994–2012)

Abstract

This chapter traces how, between 1994 and 2012, Internet platforms converted users' behaviors, relationships, and cultural production into "raw material for extraction" while providing free services. The narrative unfolds along six turning points: the establishment of the advertising-supported "free" order (§1); viral marketing turning users into transmission nodes and reducing diffusion efficiency to an equation (§2); the discovery of behavioral surplus and its conversion into sellable raw material (§3); persuasion engineering enabling platforms not merely to record behavior but to manufacture it (§4); the social graph capitalizing relationships themselves (§5); and the smartphone removing desktop constraints, making extraction an around-the-clock default (§6). A cross-cutting analysis shows that in all three major legal jurisdictions, governance never touched the substance of extraction (§7). By 2012, an engineering pipeline for converting user behavior into extractable, optimizable signals was operational and had been adopted as common practice by multiple mainstream platforms; the cognitive frame of "free is good" had settled into common sense requiring no defense; and users' behavioral data, social relationships, and cultural production were not generally recognized as contributions carrying a claim to remuneration, and no general mechanism existed for sharing the resulting commercial revenues with them. What this chapter establishes are the initial conditions of the "Understanding You" phase (§8).

Keywords

behavioral surplus; viral marketing; persuasive technology; social graph; free labor; dark patterns; surveillance capitalism; mobile Internet; zero-friction extraction; governance gap

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§1 The Establishment of the "Free" Order (1994–1998)

Around 1994, the World Wide Web transformed rapidly from a research network into a commercial medium open to the public. Graphical browsers, dial-up access, and secure connections capable of carrying transactions fell into place one after another between 1993 and 1995. Navigator, released by Netscape in early 1995, ran on Mac, PC, and Unix workstations; designed for high-speed networks and secure enough to host financial transactions, it anticipated a future in which "nearly anything could be bought with a single click." What truly broke the old commercial order was its mode of distribution—giving the browser away for free: "The company further disrupted the usual order by giving its browser away for free. It was a startlingly efficient way to build a user base. One year into its existence, Netscape had six million customers." (O'Mara, 2019) (Note: Navigator 1.0 was officially released in December 1994; the free license covered personal and academic use, while commercial licenses still required payment—O'Mara here gives a simplified account of the distribution arrangement.)

"Free" thereby became the default order of the Internet, but from the outset it had two faces: a product given away free had to be monetized elsewhere, and the channel of monetization was opened almost simultaneously. On October 27, 1994, HotWired, the digital arm of *Wired* magazine, ran the banner later celebrated as the web's first banner ad—what went live at the same time were in fact advertisements from twelve brands, including AT&T, MCI, and Volvo (Digiday, 2017). The model of advertising-supported free content was thereby established. Behind HotWired's move lay a self-conscious posture: co-founder Louis Rossetto recalled that some had warned that "if you put ads online, the internet would throw up on us," to which his response was—"There is hardly an area of human activity that isn't commercial. Why should the internet be the exception?" (Digiday, 2017) "Internet exceptionalism" was breached by the advertising business model at the very first moment of the Internet's commercialization.

The second cornerstone supporting this arrangement had been laid a few months earlier, and almost no one knew of it. In June 1994, the Netscape engineer Lou Montulli invented the HTTP cookie (Note: the entry dates the invention to June; Montulli himself recalls the concept forming in the week after a July 1994 meeting about a shopping server—the month is recorded differently across sources. HistoryofInformation.com; Montulli, 2013), originally a simple mechanism for preserving state across stateless HTTP requests; its first practical use was checking whether a visitor had already been to the Netscape site. The key lies not in the technology itself but in how it entered users' lives: cookies "were accepted by default, and users were not notified of the presence of cookies" (HistoryofInformation.com). The cookie's default presence was silent—it requested no consent and announced no presence; the inventor's design intent, moreover, had been to allow only single-site sessions and prevent cross-site tracking, and the advertising networks' use of third-party embedded resources was a later departure from that design premise (Montulli, 2013). The public did not generally learn of the cookie's existence until the *Financial Times* reported on it in February 1996.

Why did an arrangement of free content, ad-based monetization, and silent tracking not meet challenges at the

moment of its establishment? Part of the answer lies in an ideology that enveloped the early Internet. At the first Hackers' Conference in 1984, Stewart Brand left behind a much-quoted remark, but people usually quote only its second half. His complete formulation is a pair of forces locked in opposition: "information wants to be expensive, because it's so valuable," but also "information wants to be free, because the cost of getting it out is getting lower all the time. So you have these two fighting against each other" (Brand, 1984, as cited in Ford & Hoskins, 2022). "Free" was never a one-way slogan but a built-in tension—and it was precisely this tension that reserved a place for the practice of providing services for free while harvesting value elsewhere. In the mid-1990s this ideology was named "the Californian Ideology" by Richard Barbrook and Andy Cameron: a political philosophy centered on achieving individual freedom "through the natural laws of technological progress and the free market," one that "promiscuously [combines] the free-wheeling spirit of the hippies and the entrepreneurial zeal of the yuppies," that offers "technological proof to a libertarian political philosophy," and that therefore forecloses on alternative futures (Barbrook & Cameron, 1995/1996, as cited in O'Shea, 2019). In such an atmosphere, to question "free" was all but to question the Internet's own promise of liberation, and so it could find no cultural foothold.

Governance was almost entirely absent. The privacy implications of cookies did not reach the two Federal Trade Commission (FTC) hearings until 1996 and 1997, by which time silent tracking had already been running as a default for two years. What this phase left behind was not any single concrete abuse but a set of default arrangements universally taken as "natural": content was free, and the price was hidden from view. Who was being recorded, what was being recorded, and how the records would be used—these questions had not yet taken shape. They would surface in acute form only in the next phase, when the behavioral data exchanged for "free" was discovered to be usable for prediction—and for sale.

§2 The Hotmail Contagion Chain: Users Become Transmission Nodes (1996–2007)

The "free" order established in §1 gave users a new role. In advertising-supported free services, users were no longer merely the ones being served—their everyday behaviors themselves began to serve as fuel for platform expansion. Hotmail provided one of the earliest successful cases of turning this into a product mechanism.

§2.1 The Crude Contagion Chain (Hotmail, 1996–1997)

Between 1996 and 1997, Hotmail appended a single line to the bottom of every email its users sent: "Get your free e-mail at Hotmail." This line turned users' everyday correspondence into a recruitment channel for the platform—every email sent was an advertising placement. In its first year and a half, Hotmail signed up over 12 million subscribers—in the account of its founding investors, it grew a subscriber base "more rapidly than any company in the history of the world … faster than any new online, Internet, or print publication ever" (Jurvetson & Draper, 1997). It was the venture capitalists Tim Draper and Steve Jurvetson who in 1997 used the term "viral marketing" to describe this Hotmail practice—attaching advertising to the emails users sent; earlier still, Jeffrey Rayport had already used the term in his 1996 *Fast Company* article "The Virus of Marketing." The point of the analogy is this: users' social networks thereby began to be treated as a transmission medium that platforms could invoke.

Yet viral marketing at this stage remained crude engineering. Hotmail's signature line was passively embedded; the platform could neither track whether a given email had led a recipient to sign up, nor attribute each act

of transmission to subsequent conversion. The mechanism of viral growth had been identified, but not yet decomposed into a process that could be optimized item by item.

§2.2 Turning Virality into an Equation (Plaxo, 2003–2007)

Plaxo completed this step. A case study by Kalyanam, McIntyre, and Masonis, based on Plaxo's internal data and interviews with management, documented this transformation. Plaxo's product was an address-book updating service—after installing it, users chose, through the "Update Contact Wizard," which address-book contacts would receive update requests; the wizard allowed both item-by-item selection and a "check all" function that maximized the number of requests. Launched with the same reliance on viral marketing, Plaxo accumulated 5 million users in three years, but the company quickly learned that "virality" did not guarantee growth: after a brief post-launch uptick, daily adoptions fell back to single digits. The turning point came here: "Given lackluster growth and a declining cash position, Plaxo used a viral growth equation concept and an experimentation process for optimizing its 'viral engine.' The equation helped explicitly identify features of the product that impact viral growth." (Kalyanam et al., 2007) The equation formalized "virality" as a computable quantity: the new users each user could bring in depended on the number of update requests he sent out (N) and the conversion rate of those requests into new users (Cr); management computed the metric V at regular intervals to monitor the momentum of the viral engine. In this simplified model, growth stalls when V is less than 1; at V equal to 1, the user base still grows linearly; only when V is greater than 1 does growth become exponential—the original text refers to V = 1 as a "magic" threshold (Kalyanam et al., 2007). Within three months of implementing the equation, Plaxo's daily new users rose to the expected 5,000 (Kalyanam et al., 2007).

By this point, users' social networks were no longer merely a passive transmission medium; they had become objects that could be decomposed, measured, and systematically optimized. But the same case study also recorded the costs of this optimization: when the viral engine ran at full throttle, growth was accompanied by a depletion of trust—"as the viral engine took off, it set off red flags among the Internet community. Criticism of Plaxo in blogs, chat forums, and press articles tarnished it as an invasion of privacy and a generator of spam." The authors even offered an analogy: Plaxo "had become like the myth of the Ebola virus, growing so aggressively that it was in danger of killing its hosts" (Kalyanam et al., 2007). Plaxo ultimately chose adjustment: the company hired a privacy officer, modified its overly aggressive default settings, and conducted outreach toward its critics—the adjustment of its growth method was triggered by external reactions, but the viral mechanism itself was retained. This tension—between expansion efficiency and user trust—would recur on every platform that depended on viral spread.

§2.3 The Moral Neutrality of the Mechanism

The mechanism of viral marketing is itself indifferent to use. Anderson and Ling's analysis of internal tobacco-industry documents shows that RJ Reynolds developed four viral marketing channels for its Eclipse cigarette: (1) exploratory interviews to motivate consumers to spread the word about Eclipse prior to market release; (2) promotional videos featuring positive feedback from test group participants to portray a "majority consensus" among triers; (3) "Tupperware"-like parties for Eclipse, where participants received samples to pass around in their social circles; and (4) the Eclipse website's bulletin board as a forum for potential users to discuss the brand "in their own words" (Anderson & Ling, 2008). What these four channels shared was that they targeted the brand's likeliest adopters, recruited informal representatives who were unaffiliated with RJR and therefore

appeared credible, and controlled the information spread about the product (Anderson & Ling, 2008). The authors' judgment is sharper still: viral techniques "may be particularly useful to promote new tobacco products such as Eclipse that have limited appeal and need a highly motivated audience of early adopters and acceptors," because such techniques help evade the mass rejection that could follow mass promotion, circumvent marketing restrictions, and allow tobacco companies to "benefit from health claims made by consumers" (Anderson & Ling, 2008).

The same mechanism—users as transmission nodes, social relationships as diffusion channels, trust as the catalyst of conversion—could carry Hotmail to tens of millions of users just as it could let a tobacco company dress a carcinogenic product up as users' "spontaneous recommendation." The mechanism does not distinguish the purposes it serves. But turning users into transmission nodes only placed people into one link of the transmission chain. The next phase posed a different question: when users' every click and every lingering moment began to be recorded and treated as extractable raw material—"behavioral surplus" here makes its entrance.

§3 The Discovery of Behavioral Surplus: From Service Improvement to Selling Predictions (2000–2003)

In §2 users became transmission nodes, but transmission remained person-to-person—users recruiting users on the platform's behalf. The real turning point occurred elsewhere: the behavioral traces users left behind while interacting with the platform were themselves found capable of detaching from the service and becoming independent raw material. Google's business-model transformation between 2000 and 2003 is the emblematic case of this turn.

The immediate trigger of the transformation was monetization pressure. After the NASDAQ crash of March 2000, Google—still without a stable source of revenue—faced a question of survival, and its answer was AdWords. In October 2000, the engineers who coded the system tested it with a text ad of their own: "Have a credit card and 5 minutes? Get your ad on Google today" (Levy, 2011). The original AdWords sold exposure by ad position, but it soon evolved: AdWords Select, introduced in February 2002, changed to payment per click, adopted the "click-through rate" as the measure of an ad, and introduced a "quality score" that belonged to Google alone (the mechanism did not yet carry that name at the time—the designation follows Levy's 2011 retrospective account; Google introduced the name "Quality Score" in 2005—Inside AdWords, 2007)—bids and quality score jointly determined ad ranking, and the biggest factor in the quality score was the system's prediction of "the percentage of times a user would click on an ad" (Levy, 2011). AdWords Select, together with the contemporaneous AOL deal, brought Google's financial crisis to an end in 2002 (Levy, 2011).

The step from selling exposure to selling clicks looked on the surface like an improvement in pricing, but in substance it changed what the data was for. To predict whether "a user would click on an ad," the system had to accumulate and analyze users' behavioral data continuously—and the volume of such data quickly exceeded what was needed to improve the search service itself. In *The Age of Surveillance Capitalism*, Shoshana Zuboff named this excess of data, beyond what service improvement required, "behavioral surplus," and defined surveillance capitalism around it: "Surveillance capitalism unilaterally claims human experience as free raw material for translation into behavioral data. Although some of these data are applied to product or service improvement, the rest are declared as a proprietary behavioral surplus, fed into advanced manufacturing processes known as 'machine intelligence,' and fabricated into prediction products that anticipate what you will do now, soon, and

later." (Zuboff, 2019)

Zuboff marks 2002 as "the watershed year during which surveillance capitalism took root." The sign of the turn was a change in Google's attitude toward behavioral surplus: data was no longer merely a byproduct of service improvement, but became "the game-changing, zero-cost asset that was diverted from service improvement toward a genuine and highly lucrative market exchange" (Zuboff, 2019). From then on, Google's reinvestment priorities would shift "from merely improving its user offerings to inventing and institutionalizing the most far-reaching and technologically advanced raw-material supply operations that the world had ever seen," while "revenues and growth would depend upon more behavioral surplus" (Zuboff, 2019). The function of data thereby shifted from two-way exchange (users trading data for services) to one-way extraction.

Zuboff ties this extraction logic directly to a Google patent. She focuses on the patent submitted in 2003 by three of the firm's top computer scientists, titled "Generating User Information for Use in Targeted Advertising." She notes that these scientists were willing—and that their inventions were able—"to overcome the friction entailed in users' decision rights"; that Google's proprietary methods enabled it "to surveil, capture, expand, construct, and claim behavioral surplus, including data that users intentionally choose not to share"; and that "no moral, legal, or social constraints will stand in the way of" finding, claiming, and analyzing others' behavior for commercial purposes—these three sentences are Zuboff's interpretation of the patent, not the patent text itself (Zuboff, 2019).

A boundary of evidence must be drawn here: Zuboff's judgment about Google's transformation rests on her analysis of public documents, patent archives, and industry reports, not on direct access to Google's internal business-decision records; "behavioral surplus" is her theoretical reconstruction of this public evidence. The patent is a public legal document, and it does preserve the inventors' own formulations (e.g., "user profile information is not always available"; user profile information "can be determined (or updated or extended) even when no explicit information is given to the system"—Bharat et al., 2003; as quoted in Zuboff, 2019); what it can show is that technical schemes of this kind were once proposed and filed for protection, not that they were deployed as described—Zuboff's theoretical reconstruction and the patent text must be attributed separately.

The discovery of behavioral surplus also gave the order of "free" a new function. Users still believed they were "paying" for the free search service with their data; but under the logic of extraction, only a small fraction of the data users contributed flowed back as service improvement, while the rest was drawn off as behavioral surplus, processed, and sold. Part of the reason users did not notice this non-equivalence was precisely the "free" ideology described in §1: having paid nothing, one does not feel that anything has been taken. By 2003, extraction had become the foundational logic of accumulation; and once extraction itself became the objective, the next incentive followed naturally: to make users generate as much behavior as possible, and behavior as predictable as possible. How to accomplish this at the product level is precisely the task of the next phase—persuasion engineering.

§4 Persuasion Engineering Enters the Product (2003–2010)

§3 made extraction the objective; the question that immediately followed was how to make users produce more, and more predictable, behavior. The answer came from a discipline that took shape in the same period—one that turned "persuasion" from an art into something that could be engineered.

Its founder was Stanford's B. J. Fogg. In the late 1990s he founded the Persuasive Technology Lab and coined a term for the field: "captology"—an acronym based on the phrase "computers as persuasive technologies," denoting "the study of computers as persuasive technologies" (Fogg, 2003). The key lies in a shift of the object of study: captology investigates "how people are motivated or persuaded when interacting with computing products rather than through them" (Fogg, 2003). The site of persuasion thereby moved from content to the interface itself.

In 2009, Fogg condensed this line of thinking into an operational model—the Fogg Behavior Model (FBM). In this model, "behavior is a product of three factors: motivation, ability, and triggers"; for a person to perform a target behavior, he or she "must (1) be sufficiently motivated, (2) have the ability to perform the behavior, and (3) be triggered to perform the behavior. These three factors must occur at the same moment, else the behavior will not happen" (Fogg, 2009). The model's practical significance lies not in its psychological depth but in its operability: it reduces "why a person does something" to three variables that can each be adjusted from the outside. A designer need not reason with users or change their beliefs; by tuning motivation on the interface, lowering the ability threshold, and scheduling the right triggers, one obtains a direct point of leverage for changing behavior—the model itself does not guarantee that every intervention succeeds. Persuasion thereby became something that could be designed into a product, item by item.

The marketing-side counterpart of this engineering mindset is the "growth hacker." The term was coined by Sean Ellis, together with Patrick Vlaskovits (Holiday, 2013), and entered industry discourse around 2010. In Ryan Holiday's summary, "A growth hacker is someone who has thrown out the playbook of traditional marketing and replaced it with only what is testable, trackable, and scalable. Their tools are e-mails, pay-per-click ads, blogs, and platform APIs instead of commercials, publicity, and money." (Holiday, 2013) The claim is that growth need not rely on external advertising budgets but should be built into the product itself—"If you want to go viral, it must be baked into your product" (Holiday, 2013): the product must carry its own reason to be shared and the means of sharing, spreading from user to user. This is the paradigmatic form of Plaxo's viral equation in §2: treating growth as an engineerable, experimentable product feature rather than external ad placement; A/B testing became the everyday instrument for optimizing these features item by item.

Nir Eyal (2014), in *Hooked*, upgraded Fogg's single-behavior model into a self-reinforcing loop—the four-step Hook cycle: Trigger → Action → Variable Reward → Investment. Eyal explicitly takes Fogg's B=MAT as his starting point (Eyal, 2014, ch. 3), but introduces variable-ratio reinforcement from behavioral psychology as the loop's core driver: uncertain rewards fascinate more than certain ones, and so users return to the product of their own accord. The Investment step has users inject value into the product (data, content, social relationships, preference settings), making the next trigger more precise and the exit cost higher. Holiday's growth hacker solved how to bring users in (acquisition); Eyal's Hook loop solved how to keep them there and bring them back again and again (retention)—together, the two gave platforms a productized methodology both for "pulling people in" and for "locking them in."

The same behavioral-design technology also has its shadow branch. In 2010 the designer Harry Brignull named this branch: "dark patterns"—using interface design to induce users to make choices against their own interests (as recounted in Mathur et al., 2019). It shares the same methods as Fogg's persuasive technology, only with the opposite aim. Nor is this a scattered phenomenon: Mathur et al.'s large-scale study of shopping websites found that in a crawl of some 11,000 shopping websites, the researchers "discovered 1,818 instances of dark patterns," which they classified into 15 types and 7 broad categories (Mathur et al., 2019). In other words,

manipulative design is no isolated case—in that 2019 crawl of roughly 11,000 shopping websites, about one in ten sites showed dark patterns (the number is a lower bound on detection). Whether such design was already the industry "default" within the period covered by this chapter is something the study cannot answer; what the 2019 survey does establish is that by that year, manipulative design was an engineering practice with considerable coverage.

Persuasion engineering extended the platform's capabilities by one step: §3 enabled the platform to extract the behavior users had already produced, while persuasion engineering enabled the platform to manufacture behavior—inducing users to produce more, more predictable, and therefore more extractable signals. When this behavior-manufacturing technology met a new kind of platform whose skeleton was social relationships, the object of extraction would expand from individual behavior to social relationships themselves. That is the subject of the next phase.

§5 The Social Graph and the Capitalization of Data (2007–2012)

§4 taught platforms to manufacture behavior, but its object remained scattered individuals—a user induced to click once more, to linger a little longer. What extended the object of extraction from individual behavior to social relationships themselves was a class of platforms built on social relationships as their skeleton; Facebook is their paradigm.

§5.1 Data Collection Apparatuses (Like / Connect / Open Graph)

Facebook calls what it connects the "social graph." In Steven Levy's account, "Social graph refers to the nexus of connections people have in the real world" (Levy, 2020). Zuckerberg stressed to Levy that this graph is not the platform's creation: "We don't own the social graph … The social graph is this thing that exists in the world, and it always has and it always will … All we're doing is taking this real-world social graph that exists with real people and their real connections, and we're trying to get as accurate of a picture as possible of how those connections are modeled out." (Levy, 2020) This self-justification points precisely to the heart of the mechanism: the platform does not produce relationships; it takes relationships that already exist and has them "modeled out" into a machine-readable, callable graph; and once this graph was captured, "Facebook and all the other companies on the platform could exploit the social graph" to build a set of communication utilities (Levy, 2020).

What gradually converted this graph of relationships into extractable raw material were several apparatuses introduced between 2007 and 2012. The first was the "Like" button, enabled on February 9, 2009—a "thumbs up" icon. The day's technology press recorded its activation: users could henceforth express "Like" on photos, notes, and items across the network with a single click, and the feature was still rolling out gradually (Kincaid, 2009). The Like gave users a low-cost way to express approval, while generating for the platform a signal usable for ranking News Feed content: approval was discretized into a click, and the click was recycled as input to the ranking.

The second apparatus pushed the boundary of extraction beyond Facebook itself. Facebook Connect, introduced in 2008, let users log into other websites and applications with their Facebook accounts; Levy notes that "since you were logging in with Facebook, Zuckerberg's company would be able to monitor your activity," and that Facebook thereby got to share the information it held—not only about users who had intentionally signed up,

but also about "friends of users (who had no idea that their information was being passed on to apps that they might never have heard of, let alone signed up for)" (Levy, 2020). In 2010, Zuckerberg scaled this line of thinking up into the "Open Graph"—"just as the social graph mapped your personal network, the Open Graph would map the interests and activities of those you knew"; he "announced the system in 2010—Graph API V1, the first version" (Levy, 2020). Its showcase arrived the following year (2011): automatic sharing with partners including Spotify, Netflix, and the *Washington Post's* Social Reader—what you were listening to, watching, and reading was continuously pushed back into your social network. Like buttons distributed across third-party websites thus became standalone data collection points—Levy recounts privacy researcher Roosendaal's analysis: even when a user simply visited a web page that supported likes, Facebook would plant cookies on the visitor's browser; "when a user has no Facebook account, a separate set of data concerning individual browsing behavior can be created," and when that person later signs up, "the data can be connected to the newly established profile page" (Levy, 2020; Roosendaal, 2010). The data accumulated in this way is known as a "shadow profile" (Brandom, 2018); Levy records the internal 2006 Facebook concept as "dark profiles."

§5.2 The Naming and Quantification of Free Labor

By this point the social graph was no longer merely a map of relationships; it had become a map that simultaneously recorded preferences, activities, and interests. And the reason all this extraction was not called "extraction" in its early days is that it was filed under users' "voluntary and unwaged" activity. Tiziana Terranova had already named this kind of activity in her 2000 essay "Free Labor." She defined this activity as "free labor": "Simultaneously voluntarily given and unwaged, enjoyed and exploited, free labor on the Net includes the activity of building Web sites, modifying software packages, reading and participating in mailing lists …" (Terranova, 2000/2004). Her early quantitative evidence: by 1996, at the crest of the volunteer wave, "over thirty thousand 'community leaders' were helping AOL to generate at least $7 million a month"—and these people were paid nothing (Terranova, 2000/2004). Free labor was not a new invention of the digital age, but social platforms scaled it—liking, sharing, posting, tagging, commenting: these everyday gestures were at once the users' sociality and the platform's production.

This structure—labor that is never priced as such—received its quantification in industrial economics from Posner and Weyl. In the "data as labor" chapter of *Radical Markets*, they write: "Facebook, for example, pays out only about 1% of its value each year to workers (programmers) because it gets the rest of its work for free from us! In contrast, Walmart pays out 40% of its value in wages." (Posner & Weyl, 2018) At the industry level, they estimate that "the share of income going to labor in the largest tech companies is roughly 5–15%, lower than any industry other than extractive ones such as oil …," and infer from this that if these AI-driven companies represent the future of broader parts of the economy, we may be headed for a world where "labor's share falls dramatically from its current roughly 70% to something closer to 20–30%" (Posner & Weyl, 2018). Citing Jaron Lanier's judgment, they call this condition "fake unemployment"—jobs dry up "not because humans are not useful but because the valuable inputs they supply are treated as byproducts of entertainment rather than as socially valued work" (Lanier, as cited in Posner & Weyl, 2018).

Putting the two together, the substance of the social graph's capitalization becomes clear: the platform reclassified users' social relationships and cultural production as "non-labor"—a voluntary gift—thereby cancelling, in accounting terms, its obligation to pay for them. What users contributed was neither a purchased commodity nor employed labor, but something that, in the prevailing institutions, carried no recognized claim to remuneration

and no general mechanism of revenue sharing.

One theoretical tension needs flagging here: Terranova's diagnosis is rooted in the autonomist Marxist tradition—she regards free labor as a structural feature of digital capitalism and does not believe that "pricing data" can dissolve the relation of exploitation. Posner and Weyl's proposal is rooted in neoclassical welfare economics—they hold that once data is recognized as labor and compensated through market mechanisms, the problem is in principle solvable. The two start from the same empirical fact and arrive at opposite normative conclusions. This chapter juxtaposes the two as diagnostic tools and does not adjudicate between them here.

§5.3 The Absence of Governance

Governance barely touched this layer. What it occasionally ran into was the form of data sharing, not the substance of extraction: in 2007, Beacon made users' purchases on partner websites public to their social circles by default—in the internal debate over "opt-in or opt-out," the opt-out side won, on the grounds that "that's what Facebook is all about, sharing by default," and Zuckerberg ultimately "basically just overruled everyone" (Levy, 2020). The protests Beacon provoked changed it from on-by-default to capable of being disabled entirely within a month, and the feature itself was discontinued in 2009; but what was changed and withdrawn was a specific feature, not the underlying presumption of sharing by default. As for the vast pool of free labor behind the social graph—who owned it, whether its contributors ought to be remunerated—institutional claims had already surfaced (Terranova records AOL volunteers asking the U.S. Department of Labor in 1999 to investigate back wages), but they had not extended into a remuneration system covering the everyday contributions of ordinary platform users.

Up to this point, extraction had revolved mainly around desktop sessions. But mobile access did not begin with the smartphone—i-mode had merged Internet services into the phone back in 1999 (see the next section). The change of 2007–2012 lay in the gradual fusion of pocket devices, mobile data networks, location capability, and an app ecosystem; once the social graph migrated onto the smartphone—handheld and always online—extraction would expand from "while online" to "around the clock."

§6 The Mobile Turn: The Material Preconditions of Always-On (2007–2012)

What lifted the desktop constraint was not some new extraction technology but a change in the form of the terminal itself—the device for going online changed from the fixed desktop machine into the pocket computer, always online. This section sets out only the material preconditions of that change; how it flipped the target of extraction from "recording what you did" to "occupying and shaping your attention" is the business of the next stage, taken up in the next chapter (Chapter 27), "Understanding You."

The turn began with a single device. On January 9, 2007, Jobs unveiled the iPhone in San Francisco—a machine that fused three products into one: an iPod, a phone, and a device for accessing the Internet (Zittrain, 2008). In O'Mara's description, this was "a mobile phone unlike any before it: a sleek bar of metal and glass, no keyboard, no buttons, no antenna. It had a touch screen, a phone, and GPS." (O'Mara, 2019) (Note: the original iPhone's location capability actually relied on Wi-Fi and cell-tower triangulation; a dedicated GPS chip was not introduced until the iPhone 3G in 2008, and the body of the device did retain physical buttons. O'Mara here uses "GPS" as shorthand for its location capability.)

But the iPhone's significance went beyond the elegance of the hardware. Jobs initially had no intention of letting

it run third-party software—he "didn't want any of it gumming up the device's beautiful simplicity" (O'Mara, 2019). That position was quickly overturned: a year later (2008) the App Store went live, but Apple "remained firmly in control, approving any app before it could appear in the store, and taking a whopping 30 percent cut of the profits." (O'Mara, 2019) (Note: Apple's official terms describe a 70/30 split of the revenue from paid-app sales; O'Mara here calls it "profits.") Developers flocked in, and "There's an app for that" became an advertising slogan. The open web thereby gave way to a controlled app ecosystem policed by a single company.

Contemporaneously, Jonathan Zittrain identified this step as a directional turn. In *The Future of the Internet* (2008), he placed the iPhone within a larger judgment: the device marked a shift from "generative" technologies to "tethered appliances"—"The future is not one of generative PCs attached to a generative network. It is instead one of sterile appliances tethered to a network of control." (Zittrain, 2008) Generative technologies—the open PC and the Web—allowed "outsiders" to tinker freely, and it was precisely the serendipity of such outside tinkering that gave the world the Web, instant messaging, peer-to-peer networking, Skype, and Wikipedia. Once the App Store went live, Zittrain (2009) quickly followed with an analysis: the iPhone inverted the PC-era blacklist mechanism (users decide what to install, antivirus interposes after the fact) into a whitelist mechanism—outside developers had to register with Apple and submit to review, and only after approval could their apps enter the App Store, the public's principal channel for obtaining the software; Apple could change its mind at any time and withdraw any app, even remotely rewriting programs already installed on the phone (Zittrain, 2009). The result: a small number of platforms, Apple foremost among them, became the new gatekeepers, with unchecked authority over whose software could reach users (Zittrain, 2008, 2009). He also noted an irony: the man who pushed the device toward this "lockdown" was the same Jobs who had given the world its first open PC, the Apple II (Zittrain, 2008).

Competitors' responses accelerated the spread of this form without changing its logic. Facing the risk that its search revenue depended on the desktop, Google's answer was an open-source smartphone operating system, Android, "giving it away for free to any handset maker who wanted to use it" (O'Mara, 2019)—at once a gesture of openness and a commercial blocking move: Android's spread put a well-matched platform for Google's products on vast numbers of non-Apple devices, positioning Google to compete for mobile access. Free distribution let Android spread rapidly, and together with the iPhone it made the pocket, Internet-connected computer a global default terminal; by the time O'Mara wrote, there were roughly three billion smartphones worldwide (O'Mara, 2019).

Beyond the device, a network that was always online was also required. The material overture to mobile data access had been laid down before the smartphone: in February 1999, NTT DoCoMo of Japan launched i-mode, merging Internet services into the mobile phones already in common use—users could switch between calling and going online with a single button ("Before There Were Smartphones, There Was I-Mode", The History of the Web); on October 1, 2001, DoCoMo launched the world's first commercial W-CDMA 3G network (FOMA), making possible "wide-area wireless voice telephony, video calls, and broadband wireless data, all in a mobile environment" (HistoryofInformation.com). Japan thereby entered the state of the phone as the gateway to the Internet roughly two years ahead of other major markets; when 3G and the higher-bandwidth networks that followed spread worldwide together with the smartphone after 2007, "always online" acquired, for the first time and on a global scale, both of its material conditions at once—the terminal and the network.

With this, the material preconditions were complete: a device carried on the person, always online, locatable via the Global Positioning System (GPS), reachable by push notification. Going online changed from an act one

actively initiated and exited into a default state of presence. Governance barely registered this turn—Zittrain had warned back in 2008 that the controlled ecosystem would end up "leaving a handful of new gatekeepers in place"—but this private gatekeeping structure of "approval rights + funnel + 30 percent cut" was, for many years afterward, never recognized by public regulation as an object in need of discipline (the cross-cutting analysis of this governance gap is in §7).

Once this always-on material substrate was in place, the question shifted from "whether one could record continuously" to "what to do with continuous presence." When the object of recording changed from scattered sessions to a person's around-the-clock location, activity, and attention, the center of gravity of the mapping system could flip from "recording what you did" to "shaping what you want." This flip—the moment the discrete Leviathan truly took shape—is the subject of Chapter 27, "Understanding You"; this chapter stops here, having set out only the material preconditions that made it possible.

§7 Why Governance Failed to Catch It (Cross-Cutting, 1995–2012)

The six preceding sections examined six overlapping turns in sequence: the free order (§1) gave tracking a silent start; the contagion chain (§2) turned users into transmission nodes; behavioral surplus (§3) turned behavioral traces into sellable raw material; persuasion engineering (§4) enabled platforms to manufacture behavior; the social graph (§5) capitalized relationships themselves; and the mobile terminal (§6) extended recording and interaction around the clock. Curiously, across these eighteen years almost every step took place under the regulators' gaze, yet not one of them was halted wholesale. This section asks, across them all: why did governance fail to catch it?

The first layer of the answer: from the very beginning, the three major legal jurisdictions each understood "governing digital platforms" as a different thing, and not one of them faced "extraction" directly.

The European Union understood it as a question of rights. In 1995—the same year the web went commercial—the EU issued the Data Protection Directive (Directive 95/46/EC), establishing the protection of personal data as a fundamental right rather than a commodity freely tradable on the market. Its jurisprudential root is the personality right of the continental legal tradition: data is an extension of the person. This positioning prepared the institutional language and the legal precedents for the General Data Protection Regulation (GDPR) twenty years later. In terms of coverage, the Directive was not limited to "use": its Article 6 required personal data to be "collected for specified, explicit and legitimate purposes" and "adequate, relevant and not excessive in relation to the purposes" for which they were collected, and its Article 7 laid down the lawful bases for processing—the act of collection was already within regulation's reach. Its real weaknesses at this stage lay in commercial characterization and enforcement at the level of individual cases: the rules regulated "the processing of personal data," but never identified the pattern in which behavioral data becomes the consideration and operating asset of free services as an object of regulation in its own right; and rights talk did have limited persuasive power at the micro level—telling users "your data rights have been violated" is far less effective than making the service feel good to use.

The United States understood it as a problem of allocating liability—the allocation of liability for third-party content, that is, not the regulation of data relationships themselves. It had no comprehensive federal data protection law; what actually shaped the environment in which platforms grew were two "safe harbors": Section 230 of the Telecommunications Act of 1996 prevented platforms from being held liable as the "publisher or speaker"

of third-party content, and Section 512 of the Digital Millennium Copyright Act (DMCA, 1998) provided platforms with a copyright safe harbor conditioned on notice and takedown (for a legal analysis of Section 230, see Ghosh, 2020; the texts of both provisions are listed among the legal instruments below)—both provisions dealt with third-party content liability, and neither granted platforms any general authorization or exemption covering their own collection and use of user data. In the absence of comprehensive legislation, the day-to-day regulation of data relationships fell mainly to unilaterally drafted contracts and privacy policies: once users clicked "I agree," the platform treated this as an established exchange arrangement; FTC enforcement (see below) supplied a public channel beyond contract, but its cause of action was limited to deceptive statements. Individual users thus bore most of the defensive burden.

China had not yet, at this stage, formed an institution centered on a comprehensive personal data protection law (the Personal Information Protection Law was not enacted until 2021), but it was not without regulation: the Provisions on Regulating the Market Order of Internet Information Services, effective March 2012, already stipulated that information capable of identifying users could not be collected without users' consent, nor, once consent was given, collected beyond what was necessary for, or used outside the purposes of, the service provided; and on December 28 of the same year, the NPC Standing Committee's Decision on Strengthening the Protection of Network Information further established the principles of "lawfulness, legitimacy, and necessity" (author's translation), requiring the explicit disclosure of the purposes, means, and scope of collection and use of information, together with the consent of those whose information was collected. In the decade and more before this, the center of gravity of governance lay mainly in content and access—content censorship and the registration system for Internet content providers—so the collection of personal information long remained in a zone of thin rules and thinner enforcement; the two instruments differed in legal status—the first a ministerial regulation, the second a legislative decision of the NPC Standing Committee—and enforcement of both had only just begun in 2012; the judgment that there was "almost no regulation" of data extraction must be qualified accordingly.

The three understandings differed, yet shared the same blind spot: the EU regulated the conditions under which personal data were processed; the United States allocated third-party content liability; China managed content and access (and, by 2012 at the latest, introduced consent rules for collection)—none of the three identified as an object requiring regulation the fact that extraction itself had already become the business model of free services. This is the structural position of the "failure to catch"—regulation caught the declared form (consent forms, disclaimers, registration) but not the substance of extraction.

Why did even this layer of "form" fail? Because it rested on an assumption untenable from the start: that users would read, would understand, and would make informed choices. A large body of empirical research shows that users say they care about privacy yet do not protect it in practice—this gap between attitude and behavior is often called the "privacy paradox." But Acquisti, Brandimarte, and Loewenstein, in their review of research on privacy behavior, point out that this is not because users are irrational, but because privacy preferences themselves have three properties: people's "uncertainty" about the consequences of privacy-related behaviors; the "context-dependence" of people's concern about privacy; and "the degree to which privacy concerns are malleable—manipulable by commercial and governmental interests" (Acquisti et al., 2015)—the original text defines "malleable" as preferences "subject to influence by those possessing greater insight into their determinants." The examples they cite are precisely the choice of sharing defaults on social networks and interfaces offering "the provision of greater control," which "creates an illusion of safety and encourages greater sharing"

(Acquisti et al., 2015).

The other face of this mechanism is the thorough information asymmetry between the extracting side and the user. Ghosh puts it bluntly: "Only the internet baron has a holistic understanding of how the inherent commercialism of the digital ecosystem exploits the consumer." (Ghosh, 2020) What users face is "a complete asymmetry of knowledge and economic power" (Ghosh, 2020). When friction on the extraction side is engineered down to a single click while friction on the defense side (reading and amending thousands of words of terms, opting out item by item) is kept high, "consent" degenerates into a formality—users rationally choose not to read, because the cost of reading far exceeds any visible benefit. McDonald and Cranor once quantified this cost: for an American Internet user to read word for word the privacy policy of every website visited would take about 244 hours a year (the point estimate for word-for-word reading; the paper's concluding sentence attaches the $3,534 figure to approximately 201 hours)—about $3,534 per person in opportunity cost, roughly $781 billion nationwide (McDonald & Cranor, 2008)—a figure that is itself a measure of how high the friction on the defensive side was maintained. Ghosh further notes that the United States' long delay in federal privacy legislation stems mainly from the combination of this asymmetry with "hyperbolic discounting" and "industrial obscurantism," leaving Congress "unwilling to break from its inertia of dormancy" (Ghosh, 2020).

Governance in the same period was not wholly without action: the FTC's 2011 settlement with Google over Buzz barred the company from future privacy misrepresentations and required it to implement a comprehensive privacy program and submit to independent audits for the following 20 years; the FTC's 2012 settlement with Facebook likewise addressed broken privacy promises. But the cause of action in these enforcement cases was "deceptive statements"—what was punished was saying and not doing, not extraction itself. The settlements did impose behavioral requirements—consent procedures, internal privacy programs, independent audits—yet they did not touch the business model of trading free services for behavioral data, let alone establish any mechanism for returning the proceeds of extraction to users.

By 2012, each of the three paths had held its own ground—rights, market rules, national security—yet none had touched the substance of extraction, and none had built any channel that would systematically return the proceeds of extraction to its contributors. The differences among these three starting points would widen dramatically in the next stage as the capabilities of algorithmic platforms leapt forward, producing entirely different governance trajectories; but those developments (the EU's GDPR and Digital Markets Act, the antitrust suits in the United States, China's platform rectification) belong to the story of "Understanding You" and what lies beyond, and to the supply-side contest over distribution rights, and are not pursued further in this chapter (see Chapter 27 and Chapter 29).

§8 By 2012: What Each of the Three Sides Left Behind (Handover to Chapter 27)

The seven preceding sections examined the six overlapping turns of the "Collecting You" phase in sequence, together with a cross-cutting analysis of governance. By 2012—the node at which the mobile Internet erupted, and a year in which deep learning had achieved its laboratory breakthrough (the deep neural network AlexNet winning the ImageNet image-recognition competition) though it had not yet been deployed as a standard component of platform recommendation and advertising systems—one can take a snapshot and ask: what had each of the three sides accumulated, or lost?

The supply side accumulated three things. The first is data assets: the clickstreams, search terms, social

graph data, and location traces of hundreds of millions of users worldwide had been accumulated at scale; these data were not yet fully capitalized (deep-learning-driven personalized recommendation would not become a standard component of platform systems until several years later), but their scale alone already constituted a towering competitive barrier—a latecomer could hardly rebuild, from zero, more than a decade of behavioral traces. The second is infrastructure: data centers, submarine cables, Content Delivery Networks (CDNs), mobile operating systems, and app stores had formed a global physical substrate; the sunk costs of this infrastructure formed a second barrier to entry (this chapter treats infrastructure only in summary, not as a dedicated argument). The third is a methodological pipeline: from the Hotmail signature line to the Plaxo viral equation (§2), from the AdWords quality score to Google's behavioral surplus patent (§3), from the Fogg Behavior Model to dark patterns (§4), from Facebook's Open Graph to data collection points distributed across third parties (§5), and on to the App Store's controlled ecosystem (§6)—an entire engineering pipeline for converting user behavior into extractable, optimizable signals had been run through and adopted as common practice by multiple mainstream platforms. Extraction was no longer an expedient; it was an established business model.

What the user side lost is not easy to name in a single word. The free order of §1 placed users inside a cognitive frame in which, having paid nothing, one did not feel that anything was being taken away; the contagion chain of §2 turned users' everyday sociality into a recruitment channel for the platform; behavioral surplus in §3 turned users' online traces into sellable raw material detached from the service; persuasion engineering in §4 let platforms not merely record behavior but manufacture it; the social graph in §5 capitalized relationships themselves, reclassifying cultural production as an unwaged gift; and the mobile turn in §6 removed the physical constraint of the desktop terminal, making all of the above an around-the-clock default. The cumulative result: users, while continuing to receive services, had also become "raw material for extraction"—and this face of the arrangement was wholesale concealed by the narrative of "free." The behavioral data, social relationships, and cultural production that users contributed were recognized neither as commodity nor as labor; in the prevailing institutions they carried no recognized claim to remuneration, and no mechanism existed for sharing in the revenues they helped generate. By 2012, this vast store of value still had nowhere to be claimed back.

What the governance side left behind was a structural gap. §7 has already made the cross-cutting case: the EU blocked by way of a rights framework, the United States let things pass by way of market self-regulation and platform immunity, China substituted state capacity for individual protection—each path held its ground, yet none identified as an object requiring regulation the fact that extraction itself had already become the business model of free services. "Informed consent" failed structurally between zero-friction extraction and high-friction defense—McDonald and Cranor's (2008) estimate that reading all privacy policies word for word would take about 244 hours a year (the point estimate) is itself a measure of this failure—and not one institutional channel yet built systematically returned the proceeds of extraction to its contributors.

Of these three legacies, the most durable lies neither in databases nor in statutes, but in users' heads. "Free is good," "the Internet is freedom," "tech companies are different from traditional firms"—by 2012, this narrative had sedimented from a proposition in need of defense into common sense that needed none. Data can be deleted; infrastructure can be regulated; but a narrative internalized as common sense is extremely hard to dismantle—because dismantling it would mean admitting that the "voluntary" participation of the previous eighteen years had in fact taken place within a carefully engineered set of default arrangements, and that is a kind of self-correction that is cognitively very difficult.

All of this constitutes the initial conditions of Chapter 27. The "Collecting You" phase already contained pre-

diction and behavioral intervention—viral marketing changed propagation behavior, persuasion engineering manufactured behavior, click-through rate prediction entered pricing (§2–§4). What changes in the next phase is not the first crossing from "recording" to "shaping," but the granularity, frequency, and coverage of intervention: when real-time feedback becomes the norm, and personalized recommendation upgrades from one implementation among many existing techniques to a deep-learning-driven standard component, the target of extraction shifts from "recording what you did" to "occupying and shaping your attention"—from supplying information to occupying consciousness. Collection is the precondition. "Understanding You" is about what can be done with that precondition.

Placing this cross-section back into the coordinates of the whole book: what "Collecting You" accomplished was the primitive accumulation of readings in the Clock Room—vast numbers of readings were silently recorded, yet no sustained feedback loop yet existed for turning them back into operations on the clock faces; the fiber of Chapter 4 here appears as an artificial shrinking of the perceived fiber ("free is good" reads extraction as welfare rather than extraction); and the accounting discipline of Chapter 3 indicates that this liability—composed of behavioral surplus, social relationships, and cultural production—has yet to enter any double-entry ledger: it accumulates without being reconciled, and that is precisely why the next chapter, "Understanding You," can draw on it.

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